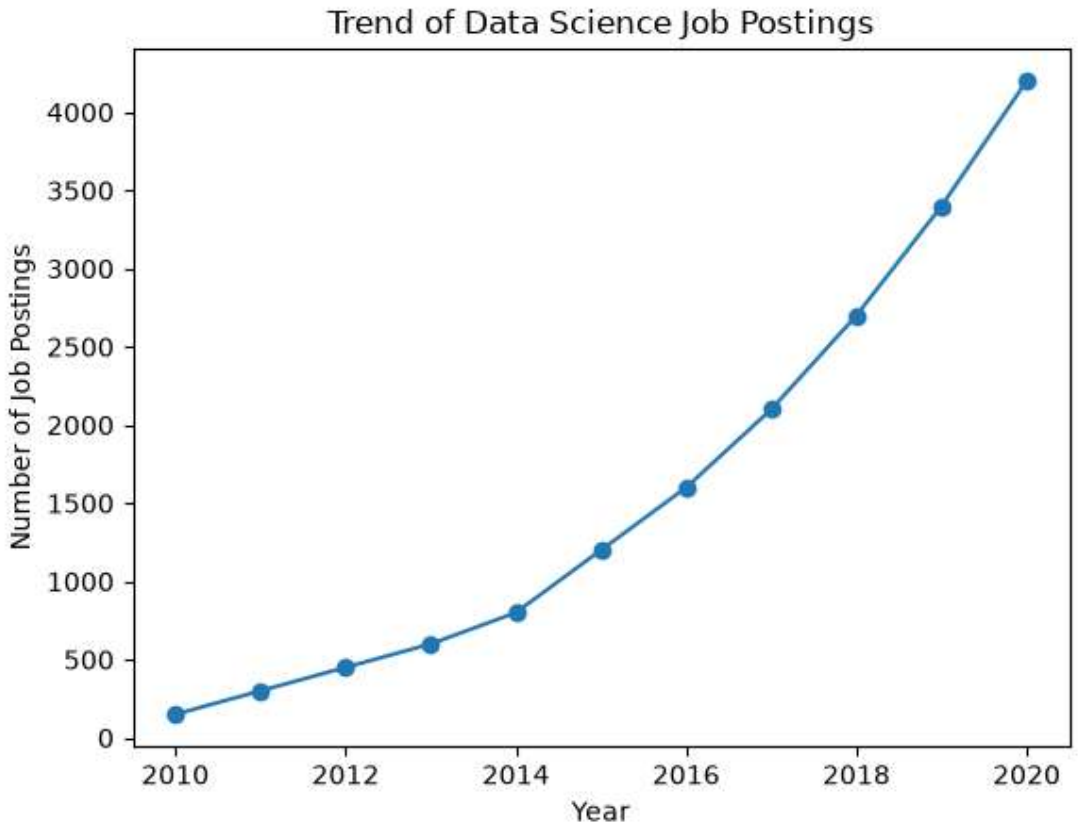
In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt

# Sample data creation
data = {
    'Year': list(range(2010, 2021)),
    'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400,
4200]
}

df = pd.DataFrame(data)

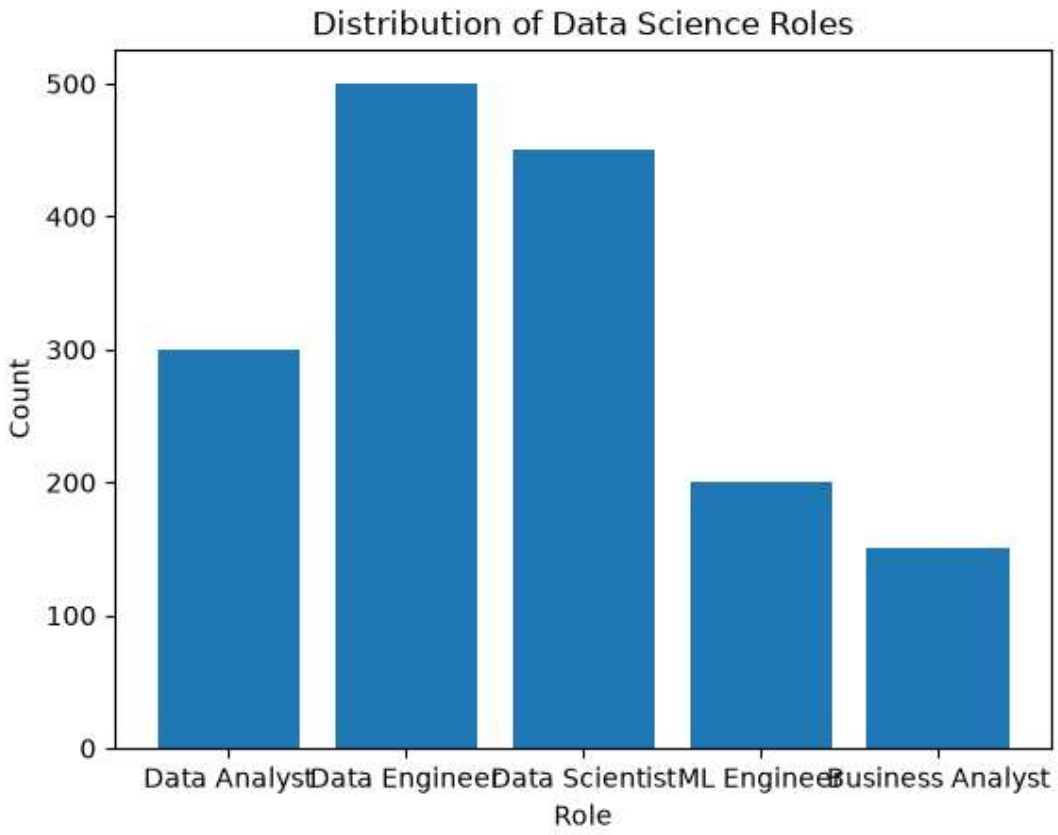
# Visualization
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```



In [2]:
```
import matplotlib.pyplot as plt

roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']
counts = [300, 500, 450, 200, 150]

plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```



In [3]:
```
import pandas as pd

# Structured data example
structured_data = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)

# Unstructured data example
unstructured_data = "This is an example of unstructured data. It can be a
piece of text, an image, or a video file."
print("\nUnstructured Data:\n", unstructured_data)

# Semi-structured data example (JSON / Dictionary)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height':
165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

```
Structured Data:
    ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
 This is an example of unstructured data. It can be a piece of text, an
image, or a video file.

Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
```

In [6]:
```
from cryptography.fernet import Fernet

# Key Generation
key = Fernet.generate_key()
cipher_suite = Fernet(key)

plain_text = b"Rajalakshmi Engineering College."

# Encrypt data
cipher_text = cipher_suite.encrypt(plain_text)

# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)

print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABqx6-
8rpks5nOjKJs0PDGoUyQjk69KqvjVuOZWi36gq4VR69jEnNoQPRue2h10beEpCMuGAl4HnY_x2Qm
hhfPEscCLj4mALFiH6J8AmxJgPT2VmFVqJ4A2APC-f1S_rtCwTEbs'
Decrypted Data: b'Rajalakshmi Engineering College.'
```